Gradient Descent for Multiple Regression

	Prev Notation	Vector Notation
Params	$w_1, \dots, w_n$ | $b \rightarrow$ a number	$\swarrow$ vector of length n $\vec{w} = [w_1 \dots w_n]$
Model	$f_{\vec{w},b}(\vec{x}) = w_1x_1 + \dots + w_nx_n + b$	$f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$ (dot product)
Cost function	$J(w_1, \dots, w_n, b)$	$J(\vec{w}, b)$

Gradient Descent	repeat {
	$w_j = w_j - \alpha \frac{\partial J(w_1, \dots, w_n, b)}{\partial w_j}$
	$b = b - \alpha \frac{\partial J(w_1, \dots, w_n, b)}{\partial b}$
	}
		repeat {
		$w_j = w_j - \alpha \frac{\partial J(\vec{w}, b)}{\partial w_j}$
		$b = b - \alpha \frac{\partial J(\vec{w}, b)}{\partial b}$
		}

Gradient Descent with one feature was,

repeat {

$$w_j = w_j - \alpha \underbrace{\frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)}\right) x^{(i)}}_{\frac{\partial J(w,b)}{\partial w}}$$

$$b = b - \alpha \underbrace{\frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}(x^{(i)}) - y^{(i)}\right)}_{\frac{\partial J(w,b)}{\partial b}}$$

simultaneously upate w, b

}

Gradient Descent with n features ($n \geq 2$):

repeat {

$j = 1$ $\quad w_1 = w_1 - \alpha \underbrace{\frac{1}{m} \sum_{i=1}^{m} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right) x_1^{(i)}}_{\frac{\partial}{\partial w_1} J(\vec{w},b)}$

$j = n$ $\quad w_n = w_n - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right) \cdot x_n^{(i)}$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{\vec{w},b}(\vec{x}^{(i)}) - y^{(i)}\right)$$

simultaneously update w_j (for $j = 1, \dots, n$) & b

}

Side note:

Alternate to Gradient Descent:

→ **Normal equation:**

- Only for linear Regression
- Solve for w & b without iterations.

Disadvantages:

- Doesn't generalize to other learning algos.
- quite slow when n is large.
  ($n > 10{,}000$)

What you need to know :

Normal equation Method may be used in machine learning libraries that implement linear regression.

- → Don't worry about detail, just know some ml libraries may use this complicated method in backend to solve for w & b.

- Gradient Descent is the reccomended method for finding parameters w & b.